

Housing layout generator — methodology and algorithm

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Purpose and scope

The generator turns a compiled site polygon into a set of candidate residential layouts — streets, plots, homes, parking, open space and a SuDS pond — and scores each one on capacity, density and profit on cost. It is a capacity and massing study, not an engineering design: it answers "roughly how many homes of what kind fit here, and what does that do to viability" in seconds, before anyone draws a line in CAD.

- **Where it lives:** `web/src/layoutgen.js` (2,275 lines), loaded on demand from the land-assembly compile panel ("Generative layout →") and from saved layouts.
- **What it produces:** plan-view geometry in local metres, rendered on the map, plus a scored summary that feeds the viability appraisal.
- **Design intent:** layouts that read as designed rather than gridded — a curved primary street from a chosen entrance, a perimeter loop on larger sites, culs-de-sac with turning heads, homes following the street curve, terraces in runs, semis in mirrored pairs, detached homes drifting to the edges and flats toward the entrance.
- **Standards basis:** tuned to Manual for Streets, Building for a Healthy Life and the Essex Design Guide. Dimensions follow those documents' typical values; nothing is checked against a statutory standard.
- **Current status:** withdrawn from the app's front end behind the `FEATURES.generativeDesign` switch (off by default); the code is intact and can be re-enabled in `web/config.js`.

Inputs and parameters

The generator takes one context object from the compile panel and exposes ten user dials; everything else is a fixed constant (see Parameter reference).

From the compile panel (`ctx`)

- `site` — the assembled site as a GeoJSON Polygon or MultiPolygon in WGS84. Several disjoint parcels are allowed; each becomes its own "part".
- `exclusions` — hard-constraint polygons already clipped to the site (flood zone, heritage, habitat). No dwelling, plot or pond may fall inside one.

- **density** , **netPct** — the target dwellings per net hectare and the net-developable share of gross, carried over from the appraisal dials (the app falls back to 35 dph and 75% when the appraisal has none).
- **assumptions** — the sift's viability variables (flat mix, tenure, costs, affordable share), and **ppm2** — the local sales value per m², both used for the profit score.
- **savedParams** — dials stored with a previously saved layout, if reopening one.

User dials (defaults in brackets)

Dial	Range	Default	What it drives
Objective	target / max / profit	Meet target density	Which score the search maximises
Target density	15-120 dph net	35 (from appraisal)	Target term of the score
Flats	0-80% of homes	20 (from flat mix)	Share of dwellings placed as flat blocks
Detached	0-80% of houses	30	Share of houses placed detached
Terraced	0-80% of houses	20	Share of houses placed in terrace runs (rest semi)
Garden min	30-150 m ²	80	Minimum rear garden; sets plot depth
Garden max	120-420 m ²	240	Cap on garden size for detached plots
Green space	0-30% of site	10	Floor on public open space
Organic	formal → organic (0-100)	70	Street curvature and plot jitter
Parking	1-3 per house	2	On-plot spaces per house; visitor spaces added

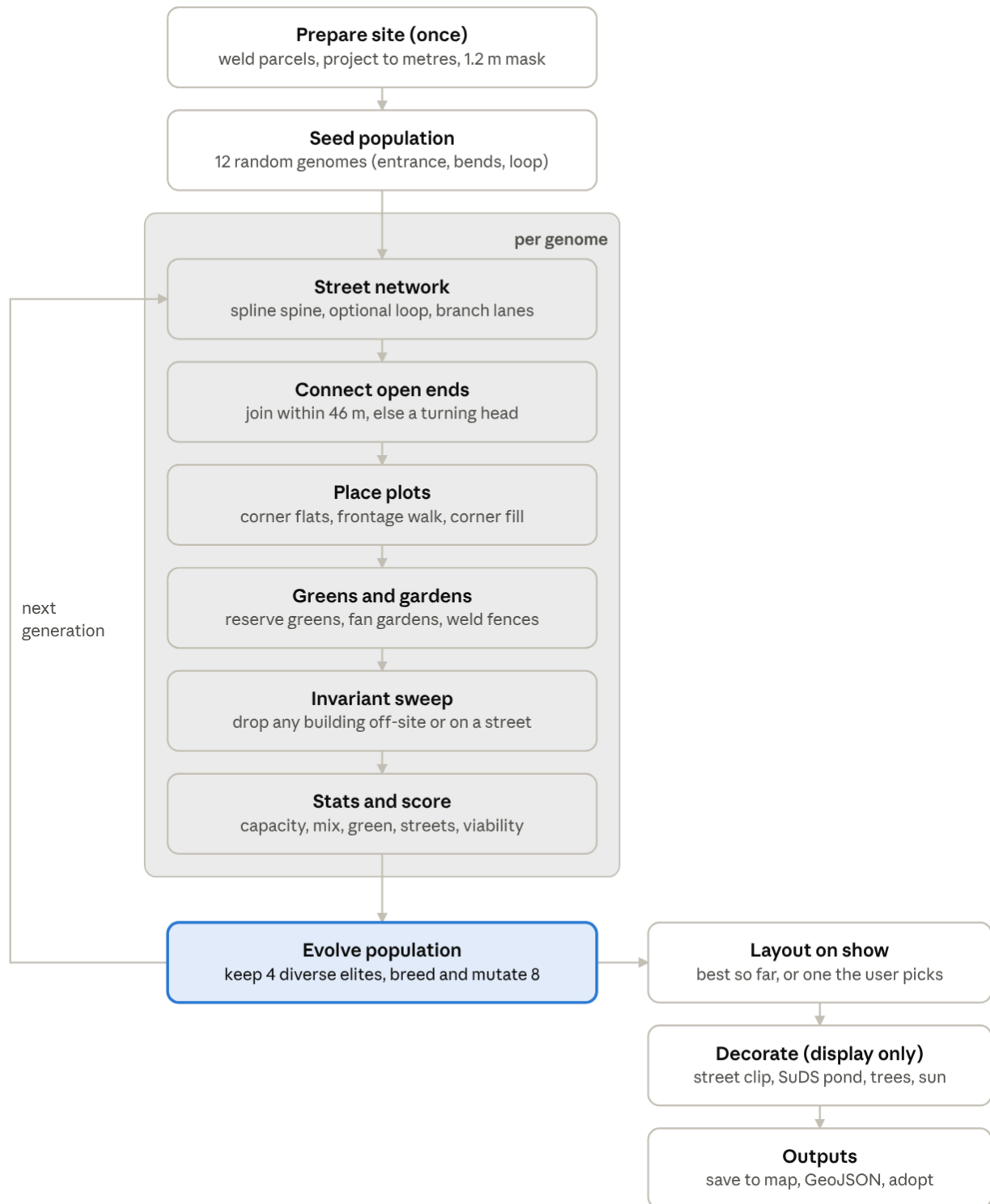
The user can also click the site edge to pin one or more **entrances**; with none pinned, the search chooses where the primary street enters.

Dials resume in priority order: the live in-memory session for the same site geometry, then dials saved with a layout, then these defaults.

Pipeline overview

The generator is a **generate-and-test search**. A deterministic procedure turns a small genome (a handful of numbers) into a complete layout; a scoring function rates it; an evolutionary loop breeds better genomes. Every stage inside the shaded box runs for every candidate, thousands of times a minute, so it is written for speed (spatial hashes, analytic areas, no polygon booleans). Anything purely visual is deferred to the one layout on show.

Each genome is built into a full layout, scored, then bred



generator pipeline · per-candidate build inside the evolution loop

Three properties follow from this design:

1. **Reproducible.** A genome carries its own random seed (Mulberry32 PRNG), so the same genome on the same site and dials always rebuilds the same layout.

2. **Rule-bound, not learned.** Every street, plot and building comes from explicit geometric rules and UK design benchmarks; there is no trained model. The search only chooses among layouts the rules can express.
3. **Hard constraints are never traded off.** Site boundary, streets and exclusion zones are enforced during placement and again by a final sweep. The score only ranks layouts that already satisfy them.

Site preparation

`prepareSite()` runs once per site and turns a messy title-parcel assembly into a clean working geometry in metres. Everything later works in this local frame.

1. **Weld.** Adjacent INSPIRE parcels rarely share exact edges, which leaves hairline slivers between them. The site is buffered out by 1.25 m and back in by 1.25 m, closing gaps narrower than 2.5 m. The welded shape is kept only if it grows the area by less than 5%; otherwise the original is used, so genuinely separate parcels are never bridged.
2. **Simplify** to about 0.2 m tolerance, removing survey noise without moving edges visibly.
3. **Project** to a local equirectangular frame centred on the site: $x = \text{longitude} \times 111,320 \times \cos(\text{latitude}_0)$, $y = \text{latitude} \times 110,540$. Over a site a few hundred metres across the distortion is well under 0.1%, far below plot tolerances, and it keeps every later distance and area test as plain arithmetic.
4. **Clean.** Holes under 30 m² are filled and fragments under 80 m² dropped; neither could hold a dwelling.
5. **Split into parts.** Each disjoint polygon becomes a *part* with its own boundary resampled every 5 m, centroid, width, diagonal and area. The largest part is flagged *main*; it hosts the primary entrance.
6. **Build a membership bitmap.** A 1.2 m grid records in-site / out-of-site for every cell. Cells within a dilated band of the boundary are flagged uncertain, and only there does `inSite(x, y)` fall back to the exact point-in-polygon test. This makes the most frequent query in the generator (tens of thousands per candidate) close to constant time while staying exact at the edge.
7. **Clip exclusions.** Flood, heritage and habitat polygons are intersected with the site to give `exclusionPolys`, their total area, and an `inExcl(x, y)` test used everywhere a street, plot, green or pond is placed.

The prepared site is cached by a hash of its geometry, so reopening the tool on the same assembly resumes instantly. Terrain (see *Open space, SuDS, parking and landscape*) loads in the background afterwards and never blocks the search.

Street network generation

Streets come first and everything else is fitted to them, mirroring how a masterplan is drawn: access and movement set the block structure, and plots follow the frontages.

Street hierarchy

Three street types, sized from Manual for Streets and typical adoptable-road practice:

Type	Carriageway	Full corridor	Used for
Primary	5.5 m	9.5 m	The spine on the main part
Secondary	4.8 m	8.4 m	Spines on other parts, perimeter loops, connectors to pinned entrances
Lane	4.4 m	6.0 m	Branch culs-de-sac and shared-surface mews

The corridor adds footways and verges to the carriageway. Turning heads are circles of 8.6 m radius, enough to turn an 11.2 m refuse vehicle.

Entrance

The primary entrance E sits on the main part's boundary. If the user has pinned entrances, E is the boundary point nearest a pin. Otherwise it is chosen by the genome gene tE (0–1), read as a fraction of the way round the resampled boundary, so the search explores every frontage. Other parts take their access from the boundary point nearest a pin, or nearest E , so a scattered assembly reads as approached from one side.

The tool does not know where the public highway is. The user supplies that knowledge by pinning entrances; without pins, the search may place the entrance on a frontage with no real road.

Per-part network (`buildPartNetwork`)

Every disjoint part gets its own mini-network, so no outlying parcel is left unserved.

Small or thin parts (under 1,500 m² or narrower than 34 m) get a single shared-surface lane traced 3.4 m inside the boundary: a mews court, with no spine.

Larger parts get a spine:

1. **Anchors.** From E to F , the boundary point furthest from E , pulled 25% back towards the centroid C . Two control points sit 45% of the way from E to C and halfway from C to F .

2. **Bends from the genome.** The control points are offset by genes `b1x`, `b1y`, `b2x`, `b2y` (each -1 to 1) scaled by an amplitude of 14% of the part's diagonal $\times$ the *organic* dial. At *formal* the spine is nearly straight; at *organic* it can sweep.
3. **Spline.** A Catmull-Rom spline through E, c1, c2, F sampled every 6 m gives a smooth street that passes exactly through its anchors.
4. **Centre it.** Any interior point closer to the boundary than $\min(0.8 \times \text{plot depth}, 0.32 \times \text{part width})$ is pushed up to 8 m towards the centroid, then the line is smoothed with a 1-2-1 kernel; three passes. This keeps a plot depth of land on both sides, so both frontages can be built. Without it a curve hugs one edge and half the street serves nothing.
5. **Trim** the far end back while it lies within 0.55 plot depths of the boundary.

Perimeter loop (gene `loop` > 0.5 , part over 12,000 m²). A ring is offset inward from the boundary by one plot depth + half a secondary corridor + 1.5 m, smoothed over 7 points, and broken wherever it leaves the part or comes within 16 m of the spine. Surviving runs over 60 m become secondary streets. The result is a perimeter block: houses face outward to the site edge and inward to the spine.

Branch lanes (spine over 6 samples, part over 5,000 m²). Walking the spine, a lane leaves every `branchGap` metres (gene, 45–105 m; the first at 60% of that), alternating sides. Each lane steps forward 6 m at a time with a random heading wobble of up to ± 0.12 rad $\times$ *organic*, and stops when it would leave the part, come within 0.7 plot depths of the boundary, or come within 15 m of another street. Lanes under 22 m are discarded. The network is capped at 16 streets.

Connectivity pass

Building for a Healthy Life favours connected networks over culs-de-sac. Every open street end looks for another street of the same part within 46 m (and more than 6 m away). If a straight connector sampled every 4 m stays inside the part, it is added and counted as a junction. Only ends that cannot connect get a turning head, and only if their street is over 24 m long; each is counted as a dead end and penalised in the score.

Each extra pinned entrance (more than 14 m from E) gets a secondary connector to the nearest street, provided it is 10–240 m long and stays inside the site.

Street land take

Each street becomes a ribbon polygon at its corridor width (for land take) and at its carriageway width (for drawing). During search, street area is estimated analytically as the sum of ribbon areas less 42 m² per junction for overlap, capped at 60% of the site. The exact union clipped to the site is a polygon boolean that is far too slow for the loop, so it is computed only for the layout on show.

For placement tests the street samples are loaded into a 9 m spatial hash, with extra samples every 2.2 m along each segment. `onRoad(x, y)` checks only the 3×3 neighbouring cells against each sample's corridor half-width + 0.15 m, turning the most common rejection test into constant time.

In **Edit streets** mode the user can drag four handles (entrance, two bends, far end). These override the spline anchors on the main part, and the layout regenerates live around the hand-drawn street.

Blocks, plots and building placement

Dwelling types

Type	Plot frontage	Floor area (GIA)	Value multiplier	Building
Detached	12.2 m	115 m ²	1.06	2 storeys, 7.8 m eaves, 2.3 m side gaps
Semi-detached	6.7 m	92 m ²	1.00	Built in pairs sharing a party wall
Terrace	5.3 m	82 m ²	0.94	Runs of 3–5 homes
Flat	26.0 m block	58 m ² per flat	0.90	14 m deep block, 2 or 3 storeys

Plot depth

Every house plot is a front garden, a house and a rear garden, measured back from the edge of the street corridor:

- **Front garden 5.2 m**, deep enough for a 2.6×5.0 m driveway space in front of the house.
- **House depth 9.2 m**.
- **Rear garden depth** = $\max(9 \text{ m}, \text{garden-min} \div 6.7 \text{ m})$. With the default 80 m² minimum that is 11.9 m, so a full plot is about 26.3 m deep. Two facing rear gardens then give about 24 m back-to-back, against the usual 21 m privacy minimum and the Essex Design Guide's 25 m aspiration.

Flat plots are 22 m deep, holding the 14 m block, a 4 m frontage and a rear strip for bins and cycles.

Unit target and mix

In *Meet target density* mode, placement stops at

$$\text{target units} = \text{density} \times \text{site hectares} \times \text{net \%}$$

(for example $35 \text{ dph} \times 2 \text{ ha} \times 75\% \approx 53 \text{ homes}$). In *Maximise* and *Profit* modes there is no cap, and placement fills every frontage that fits.

The dials set target shares: flats take the flats %, and houses split the rest into detached %, terraced % and semi (the remainder). At each frontage position the types are tried in order of **how far each is below its target share**, so the mix self-corrects as the layout fills. Position then nudges that order:

- within 30% of the site diagonal from the entrance, flats and terraces go first, but only while still under target;
- beyond 62%, detached goes first, again only while under target.

This produces the familiar gradient of higher density at the entrance and larger homes at the quiet edges. The "only while under target" guard exists because an earlier version let position override the mix, and a 20% flats dial came out at 35%.

Flats arrive in whole blocks, so they are sized to land the mix: each block is 2 or 3 storeys, whichever brings the flat share closer to target (a block holds $\text{floor}(26 \times 14 \times \text{storeys} \div 82)$ flats, 8 or 13, using 82 m² gross per flat to allow for cores and circulation). A block that would push the share further from target is refused.

Placement passes

Plots are placed in three passes, in this order.

1. Corner flat blocks. Every point where two different streets pass within 6 m of each other is a junction candidate (duplicates within 18 m merged, then shuffled by the seed). At each junction the generator tries the four ways of pairing the two street directions and keeps corners between about 52° and 129°. It fits an L-shaped, dual-aspect block 13.5 m deep that wraps the corner, trying arms of 26 × 20 m, then 21 × 16 m, then a single 18 m wing as fallback. Storeys (2 or 3) are chosen to land closest to the remaining flat budget; blocks under 6 flats are skipped. Each block gets a parking court behind it. Flats at corners are standard urban-design practice: they turn the corner with active frontage on both streets, and they make best use of the awkward land where house plots fit least well.

2. Frontage walk at full depth. Each street is walked on both sides by true arc length, so plots stay flush to a curving street. At each position s , the plot is a quadrilateral whose front edge is tangent to the street, offset by half the corridor + 0.55 m, with the plot's

frontage width along the tangent and its depth along the normal. The candidate is accepted if `tryQuad` passes:

- its centre and every edge (sampled every 3 m or finer) are in the site, off every street, and outside every exclusion;
- it does not overlap any existing plot within 60 m, tested with the separating-axis theorem (0.25 m tolerance).

Dense edge sampling matters: testing only corners let a street thread between the corners of a deep plot.

Terraces start runs of 3–5 homes and semis start pairs; a run continues with the same type and a 0.05 m party-wall gap. Between independent plots the gap is 0.5 m plus up to 1.4 m × organic of random jitter, and the house's set-back jitters by up to $\pm 0.6 \text{ m} \times \text{organic}$. This breaks up the regimented look at higher organic settings. If nothing fits, the walk advances 2 m and any run is broken.

3. Corner-fill at reduced depth. The same walk again at 55% depth (never shallower than the front garden + house + 3.5 m). This lets houses with shallower but wider gardens fill bends and site corners that full-depth plots could not, instead of leaving obvious pockets. Flats are skipped in this pass.

Building footprints

`bldQuad()` derives each building from its plot: the frontage minus side margins (2.3 m for detached, 0.7 m for other free-standing, 0.05 m inside a run), set back behind the front garden (4 m for flats), at house depth 9.2 m or flat depth 14 m. Heights are 7.8 m for houses and 6.8 m / 9.5 m for 2- and 3-storey flats; they are used in the sun check. Corner blocks use their own L-shaped footprint and wings.

Open space, SuDS, parking and landscape

Shared greens (reserve pass)

Before gardens are allowed to grow, the generator sets aside deliberate public open space up to the green-space dial (default 10% of the site):

1. Rasterise the site onto a 5 m grid and mark each cell *open* if it is in the site, off streets, outside exclusions and not inside a plot.
2. Run a breadth-first **distance transform**: cells on the edge of open land get depth 0, and each ring inward one more. The deepest cells are the cores of the largest, most usable pockets.

3. From the deepest cores (depth ≥ 2 , so at least about 10 m from any development), grow each green outward cell by cell, only into cells of depth ≥ 1 so it fills its pocket rather than leaking down narrow gaps. Up to 3 greens, each at least 140 m².
4. Trace the outline of the cell set and smooth it with two rounds of Chaikin corner-cutting, so the green reads as a shaped space rather than a staircase.

Rear gardens (infill)

After the plot passes, each plot's two rear corners march away from the street 1 m at a time until they hit a street, the site edge, an exclusion, a reserved green, or another plot. If the other plot faces the opposite way the gap is split, so the rear fences meet in the middle; otherwise the corner stops 2 m short. Growth is capped by garden-max ($\times 1.35$ for detached, $\times 0.8$ for terraces, and 22 m² per flat for flat plots) and by 24 m. The two corners may differ by at most 7 m, keeping the fan believable, and the new rear edge is re-checked along its length, since a street can cross between two clean corners.

Fence welding. Neighbouring plots on the same side of a street, less than 6 m apart, have their shared rear corner moved to the midpoint between them if that stays clear. This closes the wedge-shaped gaps that curves open between rectangular plots, so the plan reads as a continuous run of curtilage.

The result is trapezoidal plots that fill each block, with no leftover land that looks developable but is not.

Invariant sweep

A final check samples every building's outline every 2.5 m against the site, streets and exclusions. Any building that fails is removed and the unit counts corrected. It should never trigger after the checks above; it is there as a production guarantee that no dwelling stands in a street, off-site or on constrained land.

Mop-up greens

For the layout on show only, the green-carving routine runs again with no depth threshold and a 90 m² minimum, claiming every remaining sizeable pocket as a shaped green (up to 96). This is purely presentational; the score never reads it.

SuDS pond

On sites over 1.2 ha, a 12 m grid is searched for the point furthest from the boundary, every street and every plot. If that clearance exceeds 13 m and the point is outside exclusions, an ellipse of 24 $\times$ 15 m at a random orientation is drawn there as an attenuation pond. It is indicative only: no drainage calculation is made and it does not follow the site's low point.

Street and green trees

Primary and secondary streets (not lanes) get a tree every 13 m, alternating sides, 1.6 m beyond the corridor edge, where it falls inside the site and outside every plot. Each shared green gets clusters of up to 5 trees (roughly one per 90 m²), up to 60 in total.

Parking

Parking is counted, not drawn as bays:

```
spaces = houses × parking dial (default 2) + ceil(flats × 1.25) + ceil(all homes × 0.25
visitor)
```

House spaces sit on the driveway in the 5.2 m front garden; flat spaces sit in the court behind each corner block. The count is reported but is not tested against the land actually available for bays.

Sun to gardens

A first-order BRE-style check at the equinox (solar declination 0). For each house, a point 55% of the way down the rear garden is tested hourly from 08:00 to 16:00. The sun's altitude is $\text{asin}(\cos \phi \times \cos H)$ and its azimuth follows from the hour angle H and site latitude ϕ . A ray is cast towards the sun, and the hour counts as lit unless it hits a building within 46 m and within that building's shadow reach ($\text{height} \div \tan \text{altitude}$). The panel reports the median lit hours and the share of gardens with 3 hours or more. The share of gardens facing roughly south (garden normal pointing more than 20° south of east-west) is reported alongside.

Terrain and earthworks

In the background the tool fetches an elevation grid (up to 18 × 18 points, 20–45 m spacing) from the open-meteo elevation API. It derives 1 m contours by marching squares and the mean and maximum slope over in-site cells, flagging slopes over 12%. A rough earthworks volume adds, for each street, the corridor width × length × the deviation of the ground from a 7-point smoothed long-section, and for each plot, its area × the level range across its corners ÷ 4. This is an order-of-magnitude cut-and-fill indicator. It does not influence the score. If the API is unavailable the tool carries on without terrain.

Capacity metrics and scoring

Metrics (statsFor)

Metric	Definition
Dwellings	Houses + flats placed
Gross density	Dwellings ÷ gross site hectares
Net developable %	Plot land ÷ gross site area (streets, greens and leftovers excluded)
Mix deviation	flat share – target + detached share – target + terrace share – target (semi is implied)
Green %	(site – street land – plot land) ÷ site. A residual: it counts all unplotted, unpaved land, not only the drawn greens
Street length, street per home	Sum of centreline lengths; that ÷ dwellings
Dead ends, junctions	From the connectivity pass
Average rear garden	Mean of (plot area – frontage × (front garden + house depth)) over houses
Back-to-back	2 × rear garden depth; ticked at 21 m or more
South-facing gardens	Share of houses whose garden faces roughly south
Parking	As in the previous section
GIA	Sum of unit floor areas
GDV	Sum over units of floor area × local £/m ² × sales adjustment × type value multiplier
Cost	(build + streets + per-plot costs) × 1.14
Profit on cost (PoC)	(GDV – cost) ÷ cost, excluding land

The cost build-up:

- **Build:** house GIA × house £/m² (default £1,800) and flat GIA × flat £/m² (default £2,100), both × the regional cost index.
- **Streets:** £95 per m² of street land + £60 per metre of street for services in trench.
- **Per plot:** site preparation + infrastructure allowances from the sift's assumptions (£k per plot).

- **×1.14** covers fees and contingency.

Land is excluded, so PoC is a measure of **headroom to pay for land**, not a residual land value. The full appraisal remains the job of the viability panel (see *Viability link and outputs*).

Score (score0f)

Three penalties are shared by every objective:

- **Green:** $400 \times \max(0, \text{green dial} - \text{achieved green share})$. The dial is a floor, so exceeding it costs nothing.
- **Mix:** $180 \times \text{mix deviation}$.
- **Streets:** $4 \times \max(0, \text{street per home} - 8 \text{ m}) + 7 \text{ per dead end}$. An efficient layout needs about 8 m of street per home or less; each cul-de-sac head costs as much as roughly 1.75 m of extra street per home.

The objective then sets the main term:

Objective	Score
Meet target density	$1000 - 14 \times \text{gross density} - \text{density} \times \text{net \%} - \text{mix} - \text{green} - \text{streets}$
Maximise homes	$\text{dwellings} - 2 \times \text{mix} - 2 \times \text{green} - \text{streets}$
Maximise profit	$8 \times \text{PoC \%} - \text{mix} - \text{green} - 2 \times \text{streets}$

In practice, one dwelling per hectare off the target density costs 14 points, which is about the same as a 7.8-point mix deviation or 3.5% of green shortfall. The weights were tuned by hand so that no single rule dominates; they are not calibrated against real schemes.

Hard constraints (site edge, streets, exclusions, overlaps) never appear in the score, because a layout that breaks one cannot be generated.

Search and optimisation

The search is a small **elitist genetic algorithm** over street genomes. It is a practical choice rather than a novel one: the mapping from genome to score is discontinuous (a 1 m shift can add or lose a whole terrace), so gradient methods do not apply, and the genome is small enough for evolution to cover it quickly.

Genome

Gene	Range	Controls
tE	0–1, wraps	Entrance position around the main part's boundary (ignored when an entrance is pinned)
b1x, b1y	–1 to 1	Offset of the first spine bend
b2x, b2y	–1 to 1	Offset of the second spine bend
branchGap	40–110 m (random start 45–105)	Spacing of branch lanes along the spine
loop	0–1	Perimeter loop on large parts when above 0.5
seed	integer	Seeds every random choice inside the build: branch side and wobble, run lengths, jitter, junction order

The genome describes only the street network. Plots, greens and gardens are deterministic consequences of the streets and the dials, which keeps the search space small and every candidate internally consistent.

Loop

The population is **12** layouts. Each generation:

1. **Score** every candidate, plus the best-ever layout.
2. **Pick 4 elites**, best first, each more than 0.15 apart (by `gDiff`) from those already chosen; top up with the next best if fewer than 4 qualify. The best-ever always competes, so its genes are never lost.
3. **Breed 8 offspring**, each from a random elite parent:
 - 12%: a completely random genome (immigration);
 - 28%: uniform crossover with a second elite (each gene from either parent, fresh seed), then a light mutation;
 - 60%: mutation of the parent at the current mutation power.
4. **Rank** the new population and update the best-ever.

`gDiff` measures how different two street structures are: $2 \times$ the circular distance between entrances, + $0.4 \times$ each bend's offset distance, + branch-gap difference $\div 70$, + 0.8 if one has a loop and the other does not. It keeps the elite spread across genuinely different topologies instead of four near-copies of the leader.

Mutation and stagnation

At power p (1 normally):

- entrance: with probability $5\% \times p$ jump anywhere, otherwise shift by up to $\pm 6\% \times p$ of the perimeter;
- bends: shift by up to $\pm 0.25 \times p$;
- branch gap: shift by up to $\pm 9 \text{ m} \times p$, clamped to 40–110 m;
- loop: re-roll with probability $12\% \times p$;
- seed: re-roll with probability 40%.

When the best score stops improving, the mutation power rises as $p = \min(3, 1 + \text{generations since last improvement} \div 15)$. A stuck run therefore starts trying different entrances and flipped loops, not just nudging the incumbent. Every 40 generations without improvement a **restart wave** refills the non-elite population with fresh random genomes while keeping the elites.

Run-time behaviour

- The first 4 random layouts are built immediately so the panel shows something; the rest of the population fills on idle ticks every 40 ms, so the page never freezes.
- While running, the loop steps generations back-to-back for about 110 ms (at most 60), redraws once, then yields for 30 ms. The panel shows generations per second and a sparkline of the best score, which flattens as the search converges.
- Changing any dial resets the population (debounced while dragging), since scores under the old dials are not comparable. Pinning entrances or editing streets also resets.
- The population, best layout and history are kept in memory per site, so closing and reopening the tool resumes the run.

There is no fixed stopping rule: the user pauses when the sparkline has flattened. Because the elites are kept diverse, the thumbnails usually show three or four distinct street structures to choose between, not a single answer.

Viability link and outputs

The generator reads from and writes back to the rest of MasterMapper:

In. Target density and net % come from the station/site appraisal dials; the flat mix, build costs, cost index, sales adjustment and per-plot allowances come from the sift's viability assumptions; the local £/m² comes from the site's appraisal inputs (£3,500/m² if none is available). So the profit objective optimises against the same assumptions as the appraisal.

Out. Three actions act on the layout on show:

Action	What it does
Adopt into appraisal	Sends the dwelling count and achieved flat share back to the viability panel, replacing the density-based estimate with a designed capacity
Save layout to map	Stores the layout as GeoJSON with its headline stats (dwellings, density, net developable %, flat %, GIA, green %, average garden) and the dials used, and draws it on the main map; reopening restores the dials
Export GeoJSON	Downloads a georeferenced FeatureCollection for use in GIS or CAD

The GeoJSON carries one feature per element, converted back to WGS84: `street` (the exact street land clipped to the site), `shared_green`, `plot` (with house type), `building` (type, storeys, units), `suds_pond`, and `tree` points.

The PoC in the generator is a **screening figure**, useful for ranking layouts against each other. The residual land value and the policy-compliant appraisal (affordable housing, CIL/S106, finance, profit targets) are done by the viability panel once a capacity has been adopted.

Assumptions, limitations and validation

What the generator is good for

- A defensible **capacity range** for a site in minutes, from its real boundary and constraints, instead of a flat density × area figure.
- Comparing sites and assemblies on a like-for-like designed basis.
- Seeing how the dials trade against each other: flat mix vs. green space vs. garden size vs. profit.
- A starting sketch for a designer, exportable to GIS or CAD.

What it does not model

Area	Limitation	Effect
Highway access	Does not know where the adopted road is; entrances are free unless pinned	Unpinned layouts may be accessed from an impossible frontage
Existing features	No retained buildings, trees, TPOs, rights of way, easements or utilities beyond the exclusion layers	Capacity can be overstated on constrained sites
Topography	Terrain is reported, not designed for; streets ignore gradient	Steep sites (over about 12%) will lose capacity in reality
Neighbours	No overlooking or daylight test against buildings outside the site	Edge plots may need larger set-backs
Housing standards	Fixed unit sizes; no NDSS room-by-room check, no accessible (M4(2)/M4(3)) mix, no affordable tenure split in the layout	Unit counts, not unit plans
Parking	Counted, not laid out; flat courts are nominal	Parking-heavy briefs may not fit
Drainage	The SuDS pond is indicative, not sized or sited by levels	Attenuation land may be under-allowed
Refuse and servicing	Turning heads sized for a refuse vehicle; no bin-carry distances or tracking	Minor layout changes likely
Open space	The green metric is residual land, not usable public open space against a local standard (e.g. Fields in Trust)	Green % can flatter a scheme with scrappy leftovers
Flats	Generic 2–3 storey blocks; no apartment-led, mid-rise or mixed-use forms	Unsuitable as-is for dense urban and most London sites
Costs	Screening rates; no abnormals, S106/CIL or finance in the generator's PoC	Use the viability panel for decisions
Score weights	Hand-tuned, not calibrated against delivered schemes	Treat rankings as indicative

Reliability safeguards

- **Hard invariants.** No building can be off-site, in a street or in an exclusion: dense edge sampling at placement plus a final sweep.
- **Determinism.** Every layout is reproducible from its genome and dials.
- **Graceful degradation.** Terrain, the exact street clip and decoration are wrapped so a failure leaves the core layout intact.
- **Unit tests.** Geometry primitives (`bldQuad` , point-in-polygon, ring area, SAT overlap) are exported for tests.

Validation to date

The method has been checked visually on test sites and against rules of thumb (street length per home, back-to-back distance, plot sizes). It has **not** been benchmarked against a set of consented or delivered schemes. The obvious next step is to run it on 10–20 sites with known approved capacities and compare dwellings, density and net developable %, then tune the score weights and type sizes to close the gap.

Parameter reference

Fixed constants in `web/src/layoutgen.js` . Changing any of these changes every layout.

Group	Constant	Value	Basis
Streets	Primary carriageway / corridor	5.5 m / 9.5 m	Manual for Streets, local distributor
Streets	Secondary carriageway / corridor	4.8 m / 8.4 m	Minor access road
Streets	Lane carriageway / corridor	4.4 m / 6.0 m	Shared surface / mews
Streets	Turning head radius	8.6 m	11.2 m refuse vehicle
Streets	Connection search radius	46 m	Tuned
Streets	Min. length for a turning head	24 m	Short stubs left as drives
Streets	Branch gap	40–110 m	Genome
Streets	Spine bend amplitude	14% of diagonal × organic	Tuned

Group	Constant	Value	Basis
Streets	Loop minimum part area	12,000 m ²	Tuned
Streets	Branch minimum part area	5,000 m ²	Tuned
Streets	Mews-only part	< 1,500 m ² or < 34 m wide	Tuned
Plots	Front garden	5.2 m	2.6 × 5.0 m driveway
Plots	House depth	9.2 m	Typical 2-storey house
Plots	Rear garden depth	max(9 m, garden-min ÷ 6.7 m)	Garden dial
Plots	Flat plot / block depth	22 m / 14 m	Dual-aspect block
Plots	Corner block plate depth	13.5 m	Dual-aspect
Plots	Corner angle accepted	≈ 52°–129°	Tuned
Plots	Gross area per flat (sizing blocks)	82 m ²	Includes cores and circulation
Plots	Terrace run / semi pair	3–5 / 2	Common practice
Plots	Corner-fill depth	55% of full	Tuned
Plots	Entrance / edge bias zones	< 30% / > 62% of diagonal	Tuned
Buildings	Side margin (det / other / in-run)	2.3 / 0.7 / 0.05 m	Detached reads detached
Buildings	Heights (house / 2-st flat / 3-st flat)	7.8 / 6.8 / 9.5 m	Sun check
Gardens	Growth cap	24 m; garden-max ×1.35 det, ×0.8 terr; 22 m ² per flat	Tuned
Gardens	Max corner imbalance	7 m	Believable fences
Greens	Grid cell	5 m	
Greens	Reserve: count / min seed depth / min area	3 / 2 cells / 140 m ²	Tuned

Group	Constant	Value	Basis
Greens	Mop-up: count / min area	96 / 90 m ²	Presentational
Parking	Per house / per flat / visitor	dial (2) / 1.25 / 0.25 per home	Common local standards
SuDS	Pond size / min site / min clearance	24 × 15 m ellipse / 1.2 ha / 13 m	Indicative
Trees	Street spacing / greens	13 m / $\approx$ 1 per 90 m ² , max 5 each, 60 total	Indicative
Sun	Test	Equinox, hourly 08:00–16:00, 46 m search	First-order BRE
Terrain	Grid	$\leq 18 \times 18$, 20–45 m step, 1 m contours	open-meteo API
Costs	Build (house / flat)	£1,800 / £2,100 per m ² × cost index	Sift assumptions
Costs	Streets	£95/m ² + £60/m	Screening rate
Costs	On-cost	×1.14	Fees and contingency
Costs	Fallback sales value	£3,500/m ²	When no local value
Score	Target / mix / green weights	14 / 180 / 400	Hand-tuned
Score	Street threshold / per-m / dead end	8 m per home / 4 / 7	Hand-tuned
Score	Profit multiplier	8 per PoC point	Hand-tuned
Search	Population / elites	12 / 4	
Search	Random / crossover / mutate share	12% / 28% / 60%	
Search	Elite diversity threshold	gDiff > 0.15	
Search	Max mutation power / ramp	3 / +1 per 15 stagnant generations	
Search	Restart wave	every 40 stagnant generations	
Search	Batch time slice	$\approx$ 110 ms, $\leq$ 60 generations	UI responsiveness

Group	Constant	Value	Basis
Geometry	Site mask cell	1.2 m	
Geometry	Weld buffer / max area growth	± 1.25 m / 5%	
Geometry	Street hash cell / sample step	9 m / 2.2 m	
Geometry	Plot overlap tolerance	0.25 m (SAT)	